# GM-1

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The operational characteristics of a Geiger–Müller (GM) counter were investigated using gamma (Cs) and beta (Co) sources. The count rate as a function of applied voltage exhibited a well-defined plateau region for both sources, with low plateau slopes ($\sim 0.07\%/\mathrm{V}$), confirming stable detector operation. The inverse square law was examined through distance-dependent measurements, yielding a log–log slope of $-1.301$, indicating qualitative agreement with theoretical expectations but with measurable deviations due to experimental limitations. Statistical analysis of counting data demonstrated Poisson behavior for moderate counts and convergence toward a Gaussian distribution for larger counts. The detector efficiency was found to be 3.8% for gamma radiation and 32% for beta radiation, consistent with their interaction mechanisms. These results collectively validate the fundamental principles of radiation detection, propagation, and statistical fluctuations in nuclear measurements.

## INTRODUCTION

The detection and statistical characterization of ionizing radiation form a fundamental component of experimental nuclear physics. The Geiger–Müller (GM) counter, owing to its simplicity and high sensitivity, remains one of the most widely used detectors for measuring alpha, beta, and gamma radiation. Its operation is based on gas ionization and subsequent avalanche multiplication, producing measurable electrical pulses corresponding to individual radiation events.

A key feature of GM detectors is the existence of a *plateau region* in the count rate versus applied voltage characteristic, where the counting rate becomes nearly independent of the applied voltage. This property ensures reliable operation and enables accurate measurement of radiation intensity. The determination of this plateau, along with its slope and optimal operating voltage, constitutes an essential calibration step.

In addition to detector characterization, the spatial dependence of radiation intensity provides direct verification of fundamental physical laws. In particular, the inverse square law predicts that the intensity of radiation emitted isotropically from a point source decreases as the inverse square of the distance from the source. Experimental validation of this relation is crucial for understanding radiation propagation and detector response.

Furthermore, radioactive decay and detection processes are inherently stochastic. The fluctuations in measured counts are governed by Poisson statistics, reflecting the discrete and random nature of nuclear events. For sufficiently large count rates, this distribution approaches a Gaussian form, providing a bridge between discrete and continuous statistical descriptions.

In this work, we investigate (i) the characteristic response of a GM detector and determination of its operating region, (ii) the inverse square dependence of radiation intensity, and (iii) the statistical nature of nuclear counting processes. These experiments collectively illustrate the interplay between detector physics, radiation propagation, and statistical mechanics in nuclear measurements.

## THEORY

*Geiger–Müller Detector Operation.* — A GM tube consists of a cylindrical cathode and a central anode wire filled with a suitable gas mixture. When ionizing radiation enters the detector, it produces primary ionization in the gas. Under a sufficiently high applied voltage, these primary electrons initiate Townsend avalanches, leading to a large number of secondary ionizations and a measurable pulse. The process is self-quenching due to the presence of quenching gases, which suppress continuous discharge.

The counting characteristics of a GM tube are described by the variation of count rate $N$ with applied voltage $V$. At low voltages, incomplete collection of charges leads to low counts. As the voltage increases, a plateau region is reached where each ionizing event produces a full pulse independent of the initial ionization. The slope of this plateau is defined as

$$\text{Slope} = \frac{N_2 - N_1}{N_1} \cdot \frac{100}{V_2 - V_1}, \tag{1}$$

where $N_1$ and $N_2$ are the count rates at the lower and upper bounds ($V_1$, $V_2$) of the plateau. A smaller slope indicates better detector stability.

*Background Radiation and Net Count Rate.* — Even in the absence of a radioactive source, detectors register counts due to environmental radiation (cosmic rays,

terrestrial radioactivity, and electronic noise). The net count rate is therefore obtained by subtracting the background:

$$N_{\text{net}} = \frac{N - B}{T}, \tag{2}$$

where $N$ is the total counts, $B$ is the background counts, and $T$ is the counting time.

*Inverse Square Law.* — For an isotropic point source, the intensity of radiation decreases with distance due to geometric spreading. The count rate $R$ at a distance $d$ follows

$$R \propto \frac{1}{d^2}. \tag{3}$$

This relation can be experimentally verified by plotting $R$ versus $1/d^2$, which should yield a linear dependence, or equivalently by a log–log plot:

$$\log R = -2 \log d + \text{constant}. \tag{4}$$

A slope close to $-2$ confirms the inverse square behavior.

*Statistical Nature of Counting.* — Radioactive decay and detection are random processes governed by Poisson statistics. If $\bar{N}$ is the mean number of counts in a fixed time interval, the probability of observing $N$ counts is

$$P(N) = \frac{\bar{N}^N e^{-\bar{N}}}{N!}. \tag{5}$$

A key property of the Poisson distribution is that the variance equals the mean:

$$\sigma^2 = \bar{N}. \tag{6}$$

Thus, the standard deviation is

$$\sigma = \sqrt{\bar{N}}. \tag{7}$$

For large $\bar{N}$, the Poisson distribution approaches a Gaussian distribution:

$$P(N) \approx \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(N - \bar{N})^2}{2\sigma^2}\right), \tag{8}$$

which explains the emergence of symmetric distributions in high-count experiments.

*Uncertainty in Count Rate.* — The uncertainty in the measured count rate $Z = N/T$ follows from Poisson statistics:

$$\Delta Z = \frac{\sqrt{N}}{T}. \tag{9}$$

This implies that the relative uncertainty decreases as $1/\sqrt{N}$, or equivalently, as $1/\sqrt{T}$ for a constant rate, highlighting the importance of longer counting times for improved precision.

These theoretical principles underpin the experimental analysis of GM detector characteristics, radiation intensity variation, and statistical fluctuations in nuclear counting measurements.

*Detection Efficiency of the G.M. Counter.* — The efficiency of a radiation detector quantifies its ability to register incident radiation. It is defined as the ratio of the detected count rate to the rate of particles incident on the detector. For a radioactive source emitting isotropically, only a fraction of emitted particles reaches the detector due to geometric considerations.

For a point source at a distance $D$ from the detector window, the fraction of emitted radiation intercepted by the detector is determined by the solid angle subtended by the detector window. If the detector window has an effective area $A$, the geometric factor is

$$f = \frac{A}{4\pi D^2}. \tag{10}$$

For a circular window of diameter $d$, this becomes

$$f = \frac{\pi(d/2)^2}{4\pi D^2} = \frac{d^2}{16D^2}. \tag{11}$$

If the activity of the source is $A_s$ (in disintegrations per second), the rate of particles incident on the detector is

$$R_{\text{inc}} = A_s \cdot f. \tag{12}$$

The measured count rate (corrected for background) is

$$R = \frac{N - B}{T}, \tag{13}$$

where $N$ is the recorded counts, $B$ is the background counts, and $T$ is the counting time.

The intrinsic efficiency of the detector is then defined as

$$\eta = \frac{R}{R_{\text{inc}}}. \tag{14}$$

This expression accounts for both geometric attenuation and detector response. In practice, the efficiency depends on several factors including detector geometry, window thickness, and the type and energy of radiation. For gamma sources, the efficiency is typically low due to weak interaction probabilities, whereas for beta particles it is higher owing to stronger ionization in the detector gas.

It is important to note that the above formulation assumes isotropic emission and negligible absorption between the source and detector. Deviations from these assumptions can introduce systematic errors in the measured efficiency.

## EXPERIMENTAL SETUP AND PRINCIPLE

The experiment was performed using a Geiger–Müller (GM) counting system (GC-601A/602A) with an end-window GM tube mounted on an optical bench for controlled geometry. The detector was operated using an adjustable high-voltage supply, and counts were recorded with an electronic scaler over fixed time intervals.

The GM tube consists of a cylindrical cathode and a central anode wire filled with a gas mixture. Incident radiation produces ionization, leading to avalanche multiplication in the Geiger region, where each event generates a uniform output pulse independent of initial ionization.

For detector characterization, radioactive sources were placed at a fixed distance and the count rate was measured as a function of applied voltage. Background radiation was measured separately and subtracted to obtain the net count rate. The plateau region was identified from the count rate versus voltage curve, and the operating voltage was chosen within this region.

The inverse square law was studied by varying the distance between the source and detector and recording the count rate for each position. The data were analyzed in terms of $R$ vs $d$ and $R$ vs $1/d^2$ to examine the expected dependence.

Statistical properties of radioactive decay were investigated by repeated measurements over fixed time intervals. The mean, variance, and distribution of counts were analyzed to verify Poisson statistics and its Gaussian limit for large counts.

Detector efficiency was determined using a calibrated source at a known distance. By estimating the incident radiation rate from geometrical considerations and comparing it with the measured count rate, the efficiency of the detector was obtained.

Care was taken to maintain stable operating conditions and consistent geometry throughout the measurements.

## RESULTS AND DISCUSSION

Data analysis was performed using a custom Python script available at: GM_1.py

*G.M. Characteristics.* —

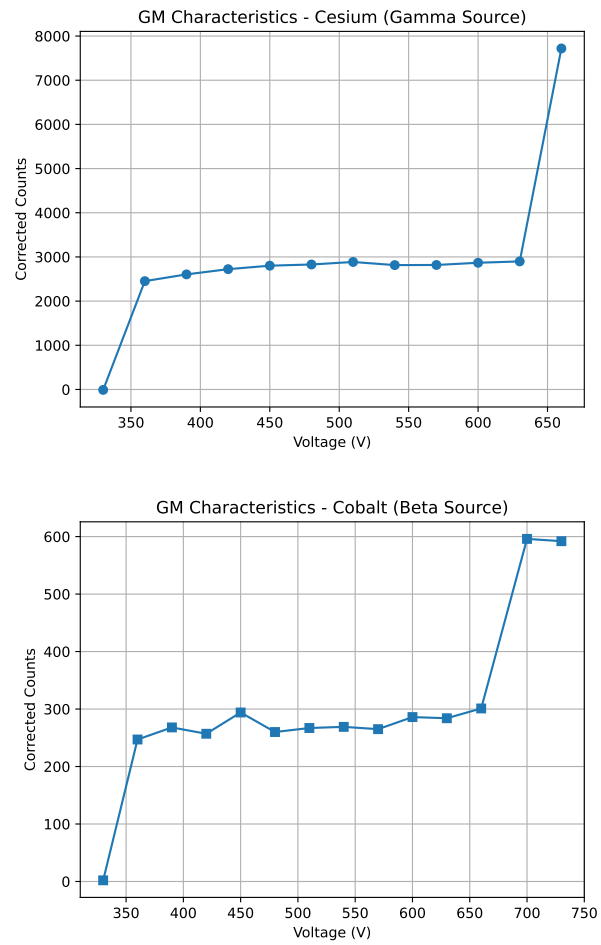


FIG. 1. Corrected count rate as a function of applied voltage for Cesium (gamma source) and Cobalt (beta source).

A well-defined plateau region is observed for both sources. For the gamma source, the plateau extends approximately from 360 V to 630 V, with count rates increasing from $N_1 = 2452$ to $N_2 = 2899$. The corresponding plateau slope is $\sim 0.068\%/\text{V}$, indicating excellent stability. The chosen operating voltage is 495 V, which lies well within the plateau.

For the beta source, the plateau extends from 360 V to 660 V, with counts increasing from $N_1 = 247$ to $N_2 = 301$. The slope is $\sim 0.073\%/\text{V}$, again indicating stable operation. The operating voltage is selected as 450 V.

The low plateau slopes in both cases confirm proper functioning of the GM detector in the Geiger region. The sharp rise beyond the plateau corresponds to continuous discharge and is excluded from operation.

*Inverse Square Law.* —

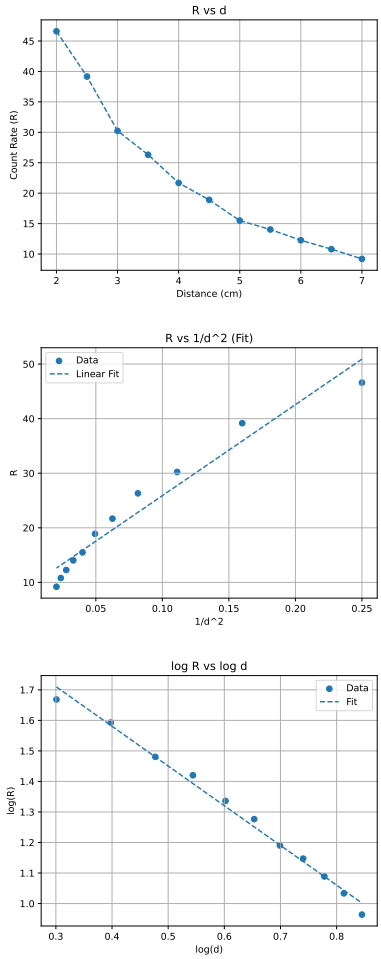


FIG. 2. (Left) Count rate vs distance. (Center) $R$ vs $1/d^2$. (Right) log–log plot.

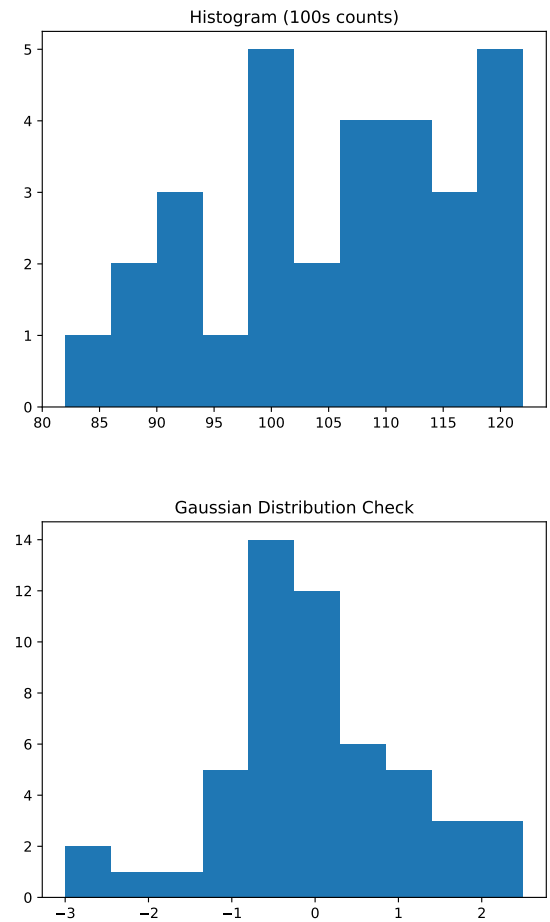


FIG. 3. Histogram of counts and normalized distribution.

The count rate decreases with distance, and the $R$ vs $1/d^2$ plot shows approximate linear behavior. The fitted relation is

$$R = 166.76\left(\frac{1}{d^2}\right) + 9.21,$$

while the log–log fit yields a slope of $-1.301$, deviating from the ideal value of $-2$. This deviation arises due to finite detector size, non-point source geometry, and scattering effects. Thus, the inverse square law is qualitatively verified.

*Statistical Behavior.* —

The statistical analysis yields $\bar{N} = 105.47$, $\sigma^2 = 114.12$, and $\sigma = 10.68$, showing that the variance is close to the mean, consistent with Poisson statistics. For larger counts ($\bar{N} = 598.12$, $\sigma = 24.46$), the distribution approaches a Gaussian form.

*Efficiency.* —

The efficiency of the GM detector was determined for both sources. The efficiency for the gamma source is found to be 0.038 (3.8%), while for the beta source it is 0.320 (32%). The significantly higher efficiency for beta radiation is consistent with its stronger ionization capability compared to gamma radiation.

Overall, the results demonstrate stable detector operation, approximate validity of the inverse square law, and statistical behavior consistent with Poisson and Gaussian distributions.

## Sources of Error

Several systematic and statistical uncertainties influence the accuracy of the measurements.

Geometrical effects constitute a primary source of systematic error. The finite size of the detector and the non-point nature of the radioactive source lead to deviations from the ideal inverse square behavior. Misalignment between the source and detector axis further contributes to discrepancies in the measured count rate.

Background radiation introduces additional uncertainty, particularly at larger distances where the signal becomes comparable to background counts. Although background subtraction was performed, fluctuations in ambient radiation can affect the net count rate.

Statistical fluctuations inherent to radioactive decay follow Poisson statistics, leading to uncertainties of order $\sqrt{N}$ in the measured counts. These fluctuations are more significant for shorter counting times and lower count rates.

Detector-related effects such as dead time and inefficiencies in charge collection can also affect the observed counts. At higher voltages, the onset of continuous discharge may artificially increase the count rate, while at lower voltages incomplete pulse formation can lead to undercounting.

Finally, uncertainties in distance measurement and timing resolution introduce additional experimental error, influencing both inverse square analysis and efficiency estimation.

## CONCLUSION

The Geiger–Müller counter was successfully characterized, exhibiting stable operation within the plateau region for both gamma and beta sources. The low plateau slopes confirmed the reliability of the detector under appropriate operating conditions. The inverse square law was qualitatively verified, although deviations from the ideal behavior highlighted the influence of finite detector geometry, scattering, and alignment uncertainties.

Statistical analysis of the counting data demonstrated clear agreement with Poisson statistics, with the expected transition to Gaussian behavior at higher count rates. The measured efficiencies, 3.8% for gamma radiation and 32% for beta radiation, are consistent with the differing interaction probabilities of these radiations in the detector medium.

Overall, the experiment illustrates the fundamental interplay between detector physics, radiation propagation, and statistical processes, providing a comprehensive validation of key principles in nuclear radiation measurements.


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